

OntologyBench: Can Dense Retrieval Satisfy Structured Biomedical Constraints?

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Abstract

We introduce OntologyBench, a tiered biomedical retrieval benchmark comprising 471,854 training and 125,744 evaluation query–document relevance pairs across concept grounding, relational retrieval, and compositional phenotype-based retrieval.

Although these tasks can be tractable using ontology-aware reference methods, across task tiers, embedding performance is generally lower on relational and compositional tasks than on concept-grounding tasks. Fine-tuning on ontology-derived supervision improves performance on several relational and compositional tasks, whereas the evaluated reranking and LLM-based candidate-scoring methods provide little or no end-to-end improvement. Errors frequently reflect diseases matching only subsets of the phenotype evidence.

These findings indicate that the evaluated embedding and reranking configurations do not reliably recover the compatibility encoded by the selected ontology relations and phenotype combinations and motivate retrieval systems that better integrate learned representations with structured biomedical knowledge.

1 Introduction

Dense retrieval models enable retrieval directly over natural language by mapping semantically related queries and documents into a shared embedding space. This supports retrieval beyond exact lexical overlap and reduces dependence on explicitly engineered symbolic methods, such as ontology-based semantic similarity scoring and graph traversal. This flexibility has made embedding retrieval attractive for biomedical and clinical

applications, where concepts are often expressed using heterogeneous, incomplete, or non-canonical language.

However, many retrieval problems require satisfying multiple interdependent constraints rather than matching individual features independently. (Agarwal et al., 2026) Existing benchmarks primarily evaluate retrieval based on unstructured semantic similarity and provide limited insight into whether models can recover compatibility across multiple structured inputs. (Muennighoff et al., 2023)

To address this gap, we introduce OntologyBench, a retrieval benchmark constructed from expert-curated biomedical ontologies. The benchmark spans concept grounding, relational retrieval, and compositional phenotype-based retrieval. These task families provide complementary views of retrieval behaviour under different forms of ontology-encoded compatibility. Because they also differ in candidate spaces, relevance structures, and query ambiguity, cross-tier comparisons are interpreted descriptively rather than as isolated estimates of constraint complexity.

Within this framework, the fixed Tier 3 phenotype-triplet-to-disease task reveals a persistent limitation. The evaluated embedding models frequently rank diseases that match only a subset of the phenotype evidence, and their performance remains substantially below that of an ontology-aware baseline. The tested rerankers provide little or no end-to-end improvement across the complete query set. These findings indicate that the evaluated retrieval systems do not reliably recover the compatibility encoded by joint phenotype sets, although

candidate omission and downstream ranking errors remain alternative sources of failure.

This work makes three contributions. First, we introduce a retrieval benchmark spanning concept grounding, ontology-relation retrieval, and phenotype-set retrieval to evaluate whether embedding models recover compatibility encoded by curated biomedical ontologies. Second, we characterize retrieval behaviour across grounding, relational, and compositional tasks, architectures, and model scales, identifying a persistent gap relative to an ontology-aware baseline on the fixed phenotype-triplet-to-disease task. Third, through two-stage evaluation, we distinguish candidate availability from the end-to-end performance of the reranked system. Across the complete query set, the evaluated rerankers provide little or no improvement over the first-stage retriever.

2 Related Work

Dense Retrieval Architectures. Dense retrieval represents queries and documents as vectors in a shared embedding space, enabling efficient large-scale search (Reimers and Gurevych, 2019). Bi-encoder models support scalable retrieval via independent encoding, while late-interaction models such as ColBERT preserve token-level representations to improve fine-grained matching (Khattab and Zaharia, 2020; Chaffin and Sourty, 2025). Recent pretrained models further improve embedding quality for retrieval tasks (Zhang et al., 2025).

Embedding Benchmarks. Large-scale benchmarks such as BEIR (Thakur et al., 2021) and MTEB (Muennighoff et al., 2023) standardize evaluation across diverse retrieval and embedding tasks, primarily measuring semantic similarity in unstructured text. Biomedical retrieval datasets such as NCorpus (Boteva et al., 2016) and TREC-COVID (Voorhees et al., 2021) similarly evaluate retrieval between clinical documents. More recent benchmarks explore retrieval requiring aggregation across multiple documents or reasoning steps, including multi-hop question answering (Yang et al., 2018) and reasoning-intensive retrieval benchmarks (Xiao et al., 2024; Su et al., 2024; Li et al., 2025). However, these benchmarks operate over unstructured text and do not isolate compositional compatibility.

Biomedical retrieval has long relied on symbolic methods that model relationships between diseases, phenotypes, and genes. Information-

content-based semantic similarity methods, including Resnik similarity and phenotype set similarity approaches used in systems such as Phenomizer, leverage hierarchical biomedical structure for retrieval. (Resnik, 1999; Köhler et al., 2009; Hoehndorf et al., 2015; Vasilevsky et al., 2025) Graph-based retrieval methods similarly exploit relational paths between biomedical entities. Unlike embedding-based retrieval, these approaches enforce compatibility through curated symbolic structure rather than learning it implicitly through representation similarity, making them less flexible for heterogeneous natural-language inputs such as free-text clinical documentation.

Recent work has also explored ontology-grounded biomedical embeddings designed to improve interpretability. QIME (Tang et al., 2026) constructs interpretable medical text embeddings by representing documents through ontology-grounded natural-language questions derived from biomedical concepts. While such approaches improve semantic interpretability and retrieval effectiveness, they primarily evaluate embedding quality through semantic similarity, clustering, and retrieval performance rather than through controlled evaluation of compositional compatibility. In contrast, OntologyBench focuses on whether retrieval systems can jointly satisfy multiple structured biomedical constraints under controlled retrieval settings.

OntologyBench evaluates whether embedding-based retrieval systems can recover the compatibility constraints enforced by symbolic biomedical retrieval methods. While embedding retrieval supports flexible natural-language matching, symbolic biomedical systems directly enforce compatibility through curated relational structure. OntologyBench organizes retrieval tasks across concept grounding, relational retrieval, and phenotype-set retrieval, enabling comparison of retrieval behaviour under different forms of ontology-defined relevance.

3 OntologyBench Dataset and Tasks

3.1 Knowledge Sources.

OntologyBench is designed as an evaluation framework spanning concept grounding, ontology-relation retrieval, and phenotype-set retrieval. The task families differ in candidate spaces, relevance structures, and query ambiguity; cross-tier comparisons are therefore interpreted descriptively. The

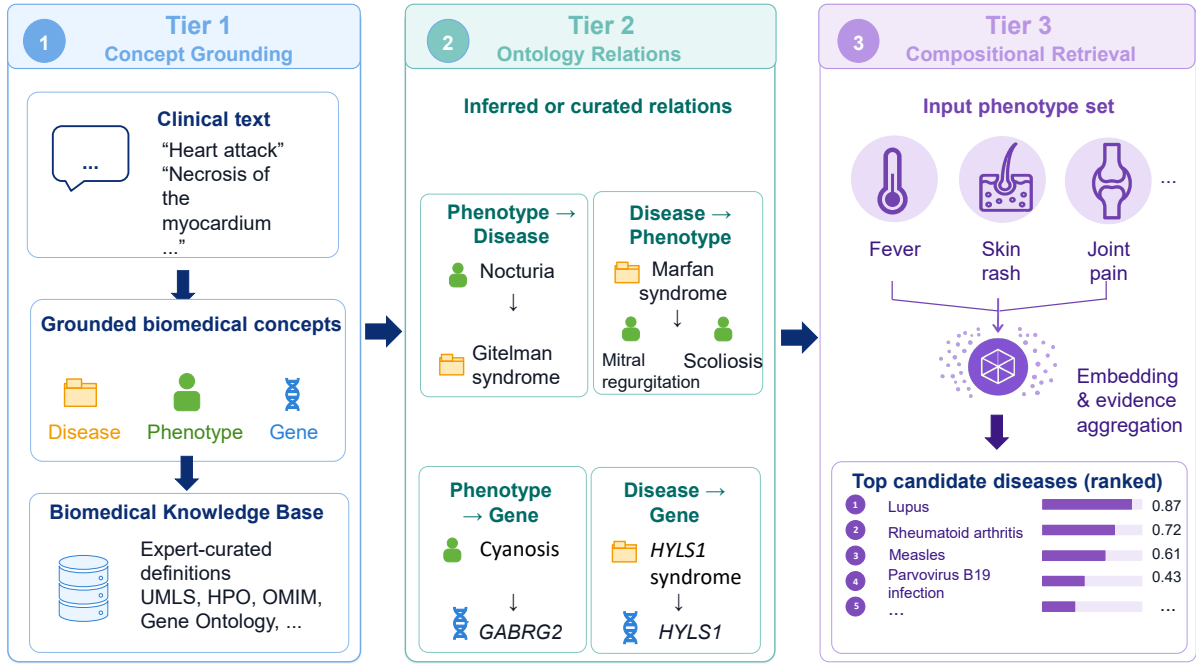


Figure 1: OntologyBench overview illustrating the retrieval tiers from grounding to compositional retrieval.

unified setting evaluates held-out directional pairs and phenotype combinations within a shared ontology vocabulary rather than generalization to entirely unseen biomedical concepts or relations.

We construct a structured retrieval space grounded in curated biomedical ontologies linking clinical findings, diseases, and genetic causes. Disease concepts are drawn from the Monarch Disease Ontology (MONDO) (Vasilevsky et al., 2025), which integrates disease terminology and mappings across resources including OMIM, SNOMED CT, and ICD. OntologyBench includes 6,611 canonical MONDO disease concepts represented by their preferred labels and available synonyms.

Phenotype concepts are drawn from the Human Phenotype Ontology (HPO) (Gargano et al., 2024), a controlled vocabulary describing observable clinical features (phenotypes). The dataset contains 8,253 phenotype concepts, each with associated textual synonyms used as query inputs.

Gene entities are constructed from HUGO Gene Nomenclature Committee (HGNC) gene names (Seal et al., 2026) combined with National Center for Biotechnology Information (NCBI) Gene metadata (O’Leary et al., 2024), resulting in 2,345 gene entries.

Together, these ontologies provide a structured biomedical knowledge space linking phenotypes, diseases, and genes, enabling evaluation of retrieval

systems under structured biomedical constraints.

3.2 Task Formulation

OntologyBench is organized into three tiers spanning concept grounding, ontology relations, and compositional retrieval (Figure 1). This organization supports comparison across concept grounding, ontology-relation retrieval, and phenotype-set retrieval. Because the task families differ in candidate spaces, relevance multiplicity, and query ambiguity, cross-tier score differences are descriptive rather than isolated effects of increasing constraint complexity. Examples of each tier can be found in Appendix A.

Tier 1: Concept Grounding Tier 1 evaluates whether models can map surface forms (aliases) of genes (Gene), phenotypes (Phen), and disorders (Dis) to their canonical ontology definitions (Def) through single-hop retrieval. This tier includes three grounding tasks: **Gene→Def**, **Phen→Def**, and **Dis→Def**.

Tier 2: Relational Retrieval. This tier evaluates whether models capture pairwise biomedical relationships between diseases, phenotypes, and genes.

Disease ↔ Phenotype (Dis↔Phen). These tasks evaluate complementary retrieval patterns over the same curated many-to-many disease–phenotype relation. Disease-to-phenotype retrieval identifies characteristic features for constructing or

reviewing disease profiles, whereas phenotype-to-disease retrieval identifies diseases compatible with an observed finding and is generally more ambiguous because phenotypes are shared across diseases. Their differing query semantics, candidate spaces, and relevance-set structures motivate evaluation in both directions.

Phenotype→Gene (Phen→Gene). Given a phenotype query, the model retrieves genes associated with diseases exhibiting that phenotype.

Disease→Gene (Dis→Gene). Given a disease query, the model retrieves genes known to be molecularly associated with that disorder.

Tier 3: PhenTriplet→Dis (compositional retrieval). In clinical settings, diagnosis is often based on incomplete and partially observed phenotypes. Tier 3 models this under a controlled phenotype-based retrieval setting, evaluating compositional retrieval from limited evidence. Each query consists of three phenotypes drawn from a single disease’s phenotype set, ensuring internal consistency. Some phenotype subsets may include non-specific features; this is intentional, as such cases introduce ambiguity and test whether models can correctly aggregate weak signals rather than rely on a single highly discriminative feature.

The choice of three phenotypes reflects a trade-off between ambiguity and determinism: single phenotypes correspond to many candidate diseases, while larger sets (≥ 4 phenotypes) sharply reduce the candidate space, making retrieval increasingly deterministic. As shown in Appendix B, phenotype triplets yield a non-trivial regime (~ 5 relevant diseases on average), requiring models to aggregate multiple constraints rather than rely on single-feature matching.

While this construction does not model the full diagnostic process, which often needs to incorporate temporal progression, demographics, disease prevalence and other contextual evidence, it intentionally isolates the compositional retrieval problem by controlling for these additional factors. This enables evaluation of whether retrieval models can integrate multiple phenotype constraints under ambiguity.

3.3 Train–Evaluation Partition and Split-Integrity Audit

The unified multi-tier corpus comprises 471,854 training and 125,744 evaluation query–document relevance pairs. Task-specific partitions were con-

Task	Train		Test	
	#Queries	AvgPos	#Queries	AvgPos
Tier 1 (Concept Grounding)				
Gene→Def	1,914	1.00	431	1.00
Phen→Def	17,763	1.00	4,581	1.00
Dis→Def	39,644	1.00	9,205	1.00
Tier 2 (Relational Retrieval)				
Dis→Phen	6,647	19.81	6,291	5.63
Phen→Dis	7,988	16.62	5,229	6.57
Phen→Gene	6,960	1.00	1,507	1.00
Dis→Gene	6,623	17.58	6,504	5.31
Tier 3 (Compositional Retrieval)				
PhenTriplet→Dis	5,642	4.39	2,537	2.26

Table 1: OntologyBench dataset statistics across tiers. AvgPos denotes the average number of positive documents per query. Tier-3 queries had an average of 4.39 relevant diseases in training and 2.26 in evaluation.

structed at the level of canonical directed query–target identifiers, and an audit of the training artifacts found zero within-task overlap of these directed pairs between training and evaluation. Because the tasks share an ontology vocabulary, concepts, targets, and underlying associations may recur across task formulations.

The two directions of Tier 2 Dis↔Phen were partitioned as separate tasks. Consequently, a disease–phenotype association evaluated in one direction could occur in training in the inverse direction. Reciprocal-task performance therefore evaluates whether shared relational supervision supports both retrieval formulations, rather than discovery of previously unseen disease–phenotype associations.

Tier 3 held out exact phenotype-triplet–disease relevance pairs and same-task positive disease targets. Most evaluation triplet queries were also absent from training; however, 146 of 2,537 queries (5.8%) occurred in training with different disease targets, and constituent phenotype–disease associations could appear in the relational training tasks. OntologyBench therefore evaluates concept grounding, transductive relational transfer, and compositional retrieval of held-out phenotype-set–disease compatibilities within a shared ontology vocabulary. It does not evaluate inductive generalization to entirely unseen entities or relation types.

3.4 Dataset Statistics

Table 1 summarizes the benchmark statistics across all retrieval tasks. AvgPos denotes the average number of ground-truth relevant (positive) target documents associated with each query. Additional dataset statistics are provided in Appendix B.

Tier 1 tasks are one-to-one concept grounding

(AvgPos = 1), while Tier 2 ontology association tasks exhibit substantial ambiguity due to many-to-many biomedical relationships (e.g., Dis→Phen and Phen→Dis average 19.8 and 16.6 positives per query). In contrast, Tier 3 compositional queries correspond to a smaller relevant set (4.39 diseases on average), reflecting the narrowing effect of combining multiple phenotype constraints. Sequence length statistics are reported in Appendix C.

4 Method

4.1 Supervision Settings

To study how ontology supervision shapes embedding representations, we compare two training regimes using the same task-specific split artifacts.

Concept Grounding Supervision. Models are trained only on alias–definition grounding pairs that map surface biomedical expressions to canonical concepts. No ontology relations or compositional signals are provided during training. Performance on higher-tier tasks, therefore, measures whether relational structure can emerge from concept grounding alone.

Unified Ontology Supervision. Models are trained jointly on all ontology-derived tasks, including grounding, entity relations (e.g., phenotype–disease, disease–gene), and compositional phenotype queries.

4.2 Two-Stage Retrieval Evaluation

Performance degradation under compositional retrieval may arise from either missing candidates during retrieval or incorrect downstream ranking among retrieved candidates. To distinguish candidate-availability failure from downstream-ranking performance, we evaluated the two stages separately. Candidate availability was measured over the complete query set.

First, an initial embedding retriever generates the top-50 candidates from the full corpus. At least one relevant disease is present among these candidates for 82.1% of queries (Hit@50 = 0.821), establishing the maximum proportion of queries for which downstream reranking can succeed. Here, Hit@k denotes the proportion of queries for which at least one relevant candidate appears among the top-k results. Second, we evaluate whether rerankers can place at least one relevant disease near the top of this fixed candidate set. Late-interaction reranking uses GTE-ModernColBERT with ColBERT-

style MaxSim scoring, whereas LLM-based candidate scoring directly evaluates phenotype–disease compatibility for each retrieved candidate (Appendix F).

Candidate-generation metrics identify queries for which reranking could not recover a relevant disease because none was available. End-to-end metrics retain all queries and therefore characterize the combined two-stage system.

4.3 Experimental Setup

Model Setup. We evaluate lexical, dense retrieval, reranking, and generative models. As a lexical baseline, we use BM25. Dense retrieval models include general-purpose embedding models such as all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), Qwen3 embeddings (0.6B and 4B) (Zhang et al., 2025), and OpenAI text-embedding-3-large¹, as well as biomedical embedding models including BioLORD-0.1B (Remy et al., 2024), MedEmbed-0.1B (Balachandran, 2024), and SapBERT-0.1B (Liu et al., 2021).

For late-interaction reranking, we evaluate GTE-ModernColBERT (Chaffin and Sourty, 2025) using ColBERT-style MaxSim scoring (Khattab and Zaharia, 2020). For LLM-based candidate scoring reranking, we use Qwen3-4B-Thinking-2507, Qwen3-4B-Instruct-2507 (Team, 2025), Qwen3.6-27B (Qwen Team, 2026), Gemma4-31B (Google DeepMind, 2026), and OpenAI GPT5.4-mini².

Ontology-Aware Reference Methods. We include ontology-aware reference methods to estimate retrieval performance when curated biomedical structure is explicitly available. These methods operate under a different information regime from text-only retrievers and are therefore interpreted as diagnostic references rather than directly comparable baselines.

For Dis→Phen, Phen→Dis, and Phen-Triplet→Dis, we use Phenomizer-style information-content similarity with set-level aggregation over HPO phenotypes. (Köhler et al., 2009) For Phen→Gene, we use graph traversal over phenotype–disease–gene paths in the train-only inductive graph. This graph contains 350 of the 447 unique Phen→Gene evaluation

¹<https://platform.openai.com/docs/guides/embeddings>

²<https://developers.openai.com/api/docs/guides/reasoning>

targets, and 1,484 of 1,507 queries can reach at least one candidate gene through an indirect path.

Note that we do not report an ontology-aware reference for Tier 1 concept grounding. These tasks require mapping textual aliases to canonical concept definitions, whereas ontology traversal requires an already grounded concept identifier. Resolving the alias through the ontology’s alias table would directly disclose the target concept and reduce the task to a tautological dictionary lookup. Also, Dis→Gene result was not reported because the same inductive graph contains none of the unique evaluation target genes. The absence of target-gene coverage prevents the method from assigning meaningful graph-derived scores. We exclude the transductive graph because it contains the held-out disease–gene edges and reduces retrieval to direct lookup.

Training Objective. Fine-tuned models use architecture-specific contrastive objectives. BioLORD-0.1B uses Cached Multiple Negatives Ranking Loss (CachedMNRL), while Qwen3-Embedding-0.6B uses InfoNCE. ModernColBERT uses the PyLate CachedContrastive objective with ColBERT-style MaxSim scoring. These objectives share the general softmax contrastive form

$$\mathcal{L}(q, d^+) = -\log \frac{\exp(\text{sim}(q, d^+))}{\sum_{d \in \mathcal{B}} \exp(\text{sim}(q, d))}, \quad (1)$$

where $\mathcal{B}$ denotes the contrastive candidate set. Similarity is computed using cosine similarity for the bi-encoder models and MaxSim aggregation for ModernColBERT. Architecture-specific loss and batching configurations are reported in Appendix D.

Training Setup. Within each architecture, the Tier 1 and unified-supervision configurations use the same epoch count and architecture-specific optimization settings. All fine-tuned configurations were trained for two epochs on one NVIDIA H100 MIG device. Complete batching, optimization, and run-accounting details are provided in Appendix D.

4.4 Evaluation Metrics

Retrieval performance is evaluated using normalized discounted cumulative gain (nDCG), mean reciprocal rank (MRR), and Hit@k, where $k \in \{1, 5, 10\}$. We report nDCG@10 in the main paper and provide additional metrics and implementation details in Appendix E. Relevance is binary,

and metrics are computed over the complete candidate set for each task and macro-averaged across queries. Although we use standard retrieval metrics, ontology structure is incorporated through task construction: queries, candidate sets, and relevance labels are derived from curated biomedical relations and compositional phenotype structures. The metrics therefore assess retrieval under ontology-constrained relevance rather than ontology consistency directly.

5 Results

5.1 Ontology-Aware Retrieval Reveals a Gap in Compositional Compatibility

Ontology-aware reference methods demonstrate that OntologyBench retrieval tasks are highly solvable when explicit biomedical structure is directly accessible. As shown in Table 3, ontology-aware methods operating over curated hierarchical and relational structure achieve strong retrieval performance across phenotype-based and compositional tasks (e.g., PhenTriplet→Dis nDCG@10 = 0.739). These ontology-aware reference methods operate under fundamentally different assumptions from embedding-based retrieval models. Rather than inferring compatibility from textual representations, they directly exploit explicit ontology hierarchy or curated disease–gene connectivity. The ontology-aware reference methods are therefore not intended as direct competitors, but instead serve as diagnostic references establishing the level of retrieval performance achievable when structured biomedical compatibility is explicitly available.

The ontology-aware reference methods reveal a substantial performance gap relative to the evaluated embedding retrieval systems. While ontology-aware reference methods directly exploit curated biomedical relationships and hierarchical structure, embedding models must recover such compatibility implicitly through representation learning from unstructured text and retrieval supervision alone. The observed performance gap suggests that current embedding training objectives may not fully preserve ontology-defined compatibility.

5.2 Retrieval Performance Differs Across Grounding, Relational, and Compositional Tasks

Table 2 shows substantial variation across retrieval tasks. Several models achieve their highest scores on concept-grounding tasks, whereas scores

Model	nDCG@10							
	Tier 1			Tier 2				Tier 3
	Gene→Def	Phen→Def	Dis→Def	Dis→Phen	Phen→Dis	Phen→Gene	Dis→Gene	PhenTriplet→Dis
BM25	0.087	0.381	0.275	0.070	0.103	0.071	0.054	0.107
MiniLM-L6-v2	0.263	0.629	0.314	0.106	0.153	0.109	0.058	0.123
MedEmbed-0.1B	0.440	0.738	0.454	0.130	0.169	0.134	0.068	0.143
SapBERT-0.1B	0.670	0.818	0.476	0.088	0.159	0.113	0.069	0.131
BioLORD-0.1B	0.290	0.807	0.426	0.091	0.178	0.105	0.058	0.151
BioLORD-0.1B [†]	0.513	0.869	0.562	0.112	0.175	0.091	0.048	0.171
BioLORD-0.1B [‡]	0.455	0.668	0.532	0.160	0.166	0.191	0.089	0.244
Qwen3-Embed-0.6B	0.325	0.605	0.247	0.089	0.151	0.106	0.063	0.109
Qwen3-Embed-0.6B [†]	0.615	0.864	0.538	0.112	0.158	0.076	0.045	0.158
Qwen3-Embed-0.6B [‡]	0.499	0.572	0.488	0.184	0.283	0.269	0.110	0.313
Qwen3-Embed-4B	0.526	0.616	0.330	0.119	0.188	0.129	0.073	0.080
OpenAI (text-embedding-3-large)	0.747	0.860	0.516	0.116	0.195	0.136	0.069	0.169

Table 2: Retrieval performance on OntologyBench, evaluated using nDCG@10 across eight retrieval tasks spanning three retrieval tiers from grounding to compositional retrieval. Fine-tuned models are highlighted in blue, where [†] indicates models fine-tuned on the Tier 1 training set only, and [‡] indicates models fine-tuned jointly on the training sets from all three tiers.

Task	Method	nDCG@10
Dis→Phen	Phenomizer-style	0.482
Phen→Dis	Phenomizer-style	0.788
PhenTriplet→Dis	Phenomizer-style	0.739
Phen→Gene	Graph traversal	0.694

Table 3: Performance of ontology-aware reference methods. Phenotype-disease and compositional tasks use Phenomizer-style ontology-based semantic similarity, while Phen→Gene uses graph traversal over phenotype-disease-gene paths.

are generally lower on relational and compositional tasks. These cross-tier differences are descriptive because candidate spaces, relevance structures, and query ambiguity also vary across tasks. On the fixed PhenTriplet→Dis task, however, the best evaluated embedding model achieves nDCG@10 of 0.313, compared with 0.739 for the ontology-aware baseline.

5.3 Task-Dependent Effects of Domain Specification and Supervision

Domain specialization provides an advantage in some comparisons but not uniformly across relational tasks. On Phenotype-Triplet→Disease, BioLORD-0.1B achieves nDCG@10 = 0.151, exceeding the pretrained Qwen3-Embed-0.6B and Qwen3-Embed-4B scores of 0.109 and 0.080, respectively.

Unified multi-tier supervision produces substantial gains on several relational and compositional tasks. Relative to its pretrained counterpart, Qwen3-Embed-0.6B improves by +0.047 to +0.163 nDCG@10 across the four Tier 2 tasks and by +0.204 on Phenotype-Triplet→Disease. It also ex-

ceeds the strongest pretrained baseline by up to +0.133 across Tier 2 and by +0.144 on Phenotype-Triplet→Disease. Some model-task combinations decline after fine-tuning, however, indicating that the benefit is not universal. Within the evaluated configurations, task-aligned supervision can produce larger gains than increasing model size, but these results do not establish a general ordering between supervision and scale. Additional evaluation results are provided in Appendix G.

5.4 Late Interaction and Generative Reranking Provide Limited End-to-End Improvement

Late-interaction rerankers substantially underperform the first-stage retriever on Tier-3 compositional retrieval tasks. Tier 1 only fine-tuning produced little improvement in ModernColBERT performance, whereas unified multi-tier fine-tuning reduced its performance (Table 4). LLM-based candidate scoring performed better than the evaluated token-interaction rerankers but the strongest LLM scorers produced only small and inconsistent differences relative to the first-stage retriever. Qwen3.6-27B increased MRR@10 from 0.297 to 0.306 and nDCG@10 from 0.313 to 0.315, while its Hit@10 was slightly lower. The strongest LLM scorers produced performance comparable to, rather than clearly exceeding, the retriever. GPT-5.4-mini produced the highest Hit@10, but its MRR@10 and nDCG@10 remained comparable to the retriever.

The first-stage retriever returned at least one relevant disease in its top-50 candidates for 2,083 of 2,537 queries (82.1%). For the remaining 454 queries (17.9%), reranking could not recover a rel-

Model	$k = 10$		
	MRR	Hit	nDCG
<i>Retriever</i>			
Qwen3-Embed-0.6B [‡]	0.297	0.572	0.313
<i>Token-Interaction Rerankers</i>			
ModernColBERT	0.168	0.384	0.180
ModernColBERT [†]	0.168	0.386	0.183
ModernColBERT [‡]	0.135	0.384	0.163
<i>LLM Candidate Scoring</i>			
Qwen3-4B-Instruct-2507	0.262	0.525	0.275
Qwen3-4B-Thinking-2507	0.278	0.546	0.288
Qwen3.6-27B	0.306	0.571	0.315
Gemma4-31B	0.284	0.524	0.287
OpenAI (GPT5.4-mini)	0.301	0.574	0.312

Table 4: End-to-end performance on the Tier-3 PhenTriplet→Dis task ($N = 2,537$). MRR@10, binary Hit@10, and nDCG@10 were calculated over the complete query set. The first-stage top-50 candidate sets contained at least one relevant disease for 2,083 queries (82.1%); the remaining 454 queries remained in the denominator and contributed zero retrieval credit. Candidate-scoring methods reordered retrieved candidates but did not introduce additional diseases.

evant disease because no relevant candidate was available to reorder. These queries were retained in the evaluation and received zero retrieval credit, so the reported metrics reflect the performance of the complete two-stage system. Because paired uncertainty estimates were not calculated, the small numerical differences among rerankers should not be interpreted as evidence that one method was superior. Overall, none of the evaluated rerankers clearly improved on the first-stage retriever, indicating that reranking did not resolve the Tier-3 retrieval bottleneck.

5.5 Failure Modes in Compositional Retrieval

We perform a rule-based error analysis (Appendix H) of Tier-3 queries for which the embedding model does not rank the ground-truth disease first. The analysis distinguishes failures for which the ontology-aware reference retrieves the ground-truth disease within the top 10 from remaining failures. Failures cluster into four modes: (1) **aggregation failure** (58.5%), where the model retrieves diseases matching only a subset of the query phenotypes rather than the full phenotype combination; (2) **generic phenotype bias** (25.0%), where retrieval is driven by broad or non-specific phenotypes; (3) **related-disease confusion** (11.8%), where the predicted disease is phenotypically similar to the ground-truth disease; and (4) **semantic drift** (4.7%), where retrieved diseases show low phenotype overlap and low ontology similarity. Ag-

gregation failures dominate, indicating that embedding models often match phenotypes independently rather than enforcing joint compatibility across the full phenotype set. Unified multi-tier supervision improves Tier 3 retrieval substantially, increasing Hit@10 from 0.170 to 0.572 and nDCG@10 from 0.109 to 0.313 (Appendix G Table 12). However, performance remains below the ontology-aware reference, indicating that compositional ranking errors remain common.

The evaluated embedding models frequently retrieved diseases compatible with only part of the phenotype set. This finding characterizes the evaluated systems and task construction; they do not establish that all embedding architectures are inherently limited to represent compositional biomedical information. This limitation is consistent with lower performance on relation types (e.g., phenotype→gene), indicating that compositional structure remains underrepresented in current representations.

6 Discussion

Retrieval systems are increasingly expected to recover compatibility across multiple structured constraints rather than simply retrieve semantically similar text. OntologyBench evaluates this distinction across concept-grounding, relational, and compositional retrieval tasks. Across the evaluated architectures, performance is generally lower on relational and compositional tasks than on several grounding tasks. These cross-task differences are descriptive because candidate spaces, relevance structures, and query ambiguity also vary. Within the fixed Tier-3 task, however, the evaluated embedding models remain substantially below the ontology-aware reference method, indicating that the evaluated text-based configurations do not recover ontology-defined relevance as effectively when multiple phenotype constraints must be combined.

Unified multi-tier supervision improves performance on several relational and compositional tasks, indicating that ontology-derived supervision can produce representations that better support these retrieval relationships. However, the remaining gap relative to ontology-aware reference methods cannot be attributed solely to the embedding objectives. The reference methods directly access curated ontology relations, whereas the embedding models operate over textual representations that

may not contain all information used to define relevance. The results therefore motivate retrieval architectures that integrate textual representations with ontology knowledge rather than establishing that embedding objectives alone are responsible for the observed gap.

Our two-stage evaluation provides additional insight into where performance is lost. Under the evaluated configurations, token-interaction reranking underperforms the first-stage retriever, while the strongest LLM-based candidate scorer provides only a modest end-to-end improvement over the fixed retriever. Because these metrics include queries for which no relevant disease appears in the initial candidate set, they reflect both candidate-generation and downstream-ranking limitations. Within the rule-selected Tier-3 failure subset, the largest category consists of predictions matching only part of the phenotype combination. Together, these findings suggest that improving downstream scoring alone may be insufficient when relevant candidates are omitted or compatibility is not adequately represented during initial retrieval.

Although OntologyBench is constructed from biomedical ontologies, its evaluation framework may be applicable beyond biomedicine. Retrieval-augmented generation, agentic systems, knowledge-graph retrieval, and other knowledge-intensive NLP applications may require candidates satisfying multiple structured constraints rather than maximizing semantic similarity alone. By organizing tasks across grounding, relational, and compositional retrieval, OntologyBench provides a framework for examining retrieval behaviour under different forms of structured compatibility. We hope it supports future work on retrieval objectives and architectures that more effectively integrate representation learning with structured knowledge.

7 Conclusion

We introduced OntologyBench, a benchmark spanning concept grounding, ontology-relation retrieval, and phenotype-set retrieval. Scores were generally lower on relational and compositional tasks than on several grounding tasks. On the fixed PhenTriplet→Dis task, the evaluated embedding configurations remained below the ontology-aware reference methods, and the evaluated rerankers provided little or no end-to-end improvement over the first-stage retriever. Candidate omission contributed to the observed end-to-end gap.

These findings motivate retrieval systems that combine flexible textual representation with explicit structured knowledge; they do not establish a universal ordering between embedding and symbolic approaches.

8 Data and Code Availability

The OntologyBench benchmark is publicly available on [GitHub](#) (v1.0.0; MIT License), with the benchmark dataset additionally distributed through [Hugging Face](#) under the CC BY 4.0 License.

Limitations

OntologyBench evaluates retrieval under controlled biomedical compatibility constraints and does not capture broader aspects of real-world diagnosis, such as temporal progression, causal mechanisms, incomplete observations, or patient-specific context.

Task-specific partitioning prevents overlap of exact query–document pairs and same-task target identifiers. However, biomedical concepts and underlying relations may recur across training tasks. The unified setting therefore evaluates transductive multi-task transfer within a shared ontology vocabulary rather than generalization to entirely unseen entities or relations. However, scores were generally lower on relational and compositional tasks than on several grounding tasks, although task characteristics differ and these cross-tier contrasts should not be interpreted as isolated effects of constraint complexity (Table 2).

Our experiments evaluate a representative range of retrieval architectures and model scales, but are not intended as an exhaustive leaderboard-style comparison of all emerging foundation systems. The primary goal of OntologyBench is instead to characterize retrieval behaviour across grounding, relational, and compositional retrieval tasks. Future work may investigate whether larger-scale or ontology-aware architectures mitigate the limitations identified here.

OntologyBench evaluates retrieval behaviour under structured biomedical constraints rather than directly measuring preservation of ontology geometry within embedding space. Standard retrieval metrics such as nDCG, MRR, and Hit@k therefore do not explicitly quantify hierarchical distance preservation or geometric alignment with ontology topology. Consequently, the lower scores observed on relational and compositional retrievals should

be interpreted as evidence of limitations in compositional retrieval rather than a definitive measure of ontology structure preservation within learned representations.

Although ontology-aligned supervision improves relational and compositional retrieval, the current study does not fully isolate which components of the supervision signal contribute most strongly to these gains. Improvements may arise from a combination of structured relational supervision, task-specific retrieval supervision, and domain adaptation effects. Future work could further disentangle these factors through more controlled supervision ablations.

Finally, OntologyBench focuses on representation-level retrieval. While practical clinical systems may incorporate reranking, generative reasoning, and expert validation, our results suggest that substantial limitations already emerge at the retrieval stage. The benchmark should therefore be interpreted as a diagnostic framework for analyzing retrieval behaviour under structured biomedical constraints rather than a direct proxy for end-to-end clinical performance.

Ethical Considerations

OntologyBench is constructed entirely from publicly available biomedical ontologies released under open licenses, including the HPO and MONDO (CC-BY 4.0) and HGNC and NCBI resources (public domain). These sources provide structured definitions of concepts and relations rather than patient-level data. OntologyBench contains no personally identifiable information or sensitive health records.

Statement of AI Use

Generative AI was used for language editing. All research design, methodology, experiments, analyses, results, and conclusions are the work of the authors and were verified by the authors.

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A Dataset Examples

This appendix shows representative examples from the three tiers of OntologyBench tasks.

A.1 Tier 1: Concept Grounding

Gene→Document Retrieval

doc_id: GENE:2011
doc: Protein-coding gene located on chromosome 11q13.1 belonging to the microtubule affinity regulating kinase family. The encoded kinase regulates epithelial and neuronal cell polarity and microtubule stability.

Phen→Definition Retrieval

doc_id: HP:0006349
doc: Congenital absence of one or more permanent teeth, including hypodontia, oligodontia, or complete tooth absence.

MONDO→Definition Retrieval

doc_id: MONDO:0014543
doc: Congenital myasthenic syndrome with glycosylation defect caused by mutations in the ALG2 gene.

A.2 Tier 2: Ontology-Level Relations

Disease→Phenotype

doc_id: HP:0002154
doc: Elevated concentration of glycine in the blood.

Phenotype→Disease

doc_id: MONDO:0009479
doc: Johanson-Blizzard syndrome characterized by exocrine pancreatic insufficiency, nasal alae hypoplasia,

hearing loss, growth retardation, and intellectual disability.

Phenotype→Gene

doc_id: GENE:80185
doc: TTI2 encodes a regulator of the DNA damage response and a component of the Triple T complex involved in cellular resistance to DNA damage stresses.

Disease→Gene

doc_id: GENE:889
doc: KRIT1 encodes a protein involved in beta1-integrin-mediated cell proliferation and endothelial junction integrity. Mutations cause cerebral cavernous malformations.

A.3 Tier 3: Compositional Phenotype-Based Disease Retrieval

Multi-Phenotype→Disease

doc_id: MONDO:0016394
doc: Sporadic infantile bilateral striatal necrosis characterized by degeneration of the caudate nucleus, putamen, and globus pallidus, leading to developmental regression and movement disorders.

Multi-Phenotype→Disease

doc_id: MONDO:0011614
doc: HMG-CoA synthase-2 deficiency, a rare disorder of ketone body metabolism presenting with vomiting, lethargy, hepatomegaly, and non-ketotic hypoglycemia.

Multi-Phenotype→Disease

doc_id: MONDO:0020502
doc: Yellow fever, a zoonotic viral disease that may progress from fever and systemic symptoms to hemorrhagic fever and multi-organ failure.

B Additional Dataset Analysis

Tier 1 Statistics

Disease identity grounding (MONDO) contains 6,611 canonical concepts associated with 48,854

alias queries (mean 7.39 aliases per concept; median 6; maximum 50).

Alias polysemy is limited: 1.0% of surface forms map to more than one concept. Gene identity grounding contains 2,345 gene concepts, each associated with a single canonical symbol. Phenotype grounding includes 8,253 phenotype concepts with extensive synonym coverage.

These statistics indicate that Tier 1 primarily evaluates lexical normalization under moderate alias diversity rather than heavy ambiguity.

Phenotype Cardinality and Hypothesis Space Reduction

We fix Tier-3 phenotype cardinality to three to study the minimal non-trivial regime of compositional retrieval. Ontology association statistics indicate:

- Single phenotype: mean 328.9 compatible diseases
- Phenotype pairs: mean 26.7 compatible diseases
- Phenotype triplets: mean 5.0 compatible diseases
- Phenotype quadruplets: mean 1.95
- Phenotype quintuplets: mean 1.31

Triplets provide substantial hypothesis-space reduction while avoiding near-deterministic collapse observed at higher cardinalities.

C Experimental Details

Sequence Length Distribution and Truncation

Model	Mean	P95	P99	Max
Qwen3-Embed-0.6B	33.40	112	161	1053
BioLORD-0.1B	48.39	128	205	926
ColBERT	25.08	88	149	845

Table 5: Token length statistics of training instances under different tokenizers (no truncation applied). We report the mean length, 95th percentile (P95), 99th percentile (P99), and maximum sequence length.

Bi-encoder models used a maximum sequence length of 256 tokens to limit truncation under the heavy-tailed definition-length distribution. ModernColBERT retained its architecture-specific maximum lengths of 48 query tokens and 300 document tokens. Optimization, batching, hardware, and run-accounting details are provided in Appendix D.

D Optimization and Training Setup

All fine-tuned configurations were trained for two epochs on one NVIDIA H100 MIG device with gradient accumulation of 1 and random seed 42. Reported fine-tuning results are from single training runs rather than averages across multiple seeds.

BioLORD-0.1B was fine-tuned with Cached Multiple Negatives Ranking Loss using a batch size of 256 and cached forward mini-batches of 128. Qwen3-Embedding-0.6B was fine-tuned with InfoNCE using a batch size of 256 packed records, sequence packing with a length of 256 tokens, and no cached-loss subdivision. ModernColBERT was fine-tuned with the PyLate CachedContrastive objective using a batch size of 256 and cached forward mini-batches of 32. Cross-device negative gathering was disabled for cached-loss configurations.

Bi-encoder models used a maximum sequence length of 256 tokens. ModernColBERT used maximum query and document lengths of 48 and 300 tokens, respectively. Learning rates were 2×10^{-5} for BioLORD-0.1B, 5×10^{-5} for Qwen3-Embedding-0.6B, and 3×10^{-6} for ModernColBERT. Complete training configurations and saved trainer states are included in the released repository.

E Evaluation Protocol

For each task, every query is evaluated by ranking all documents in the corresponding task-specific corpus without heuristic filtering or ontology pruning. Metrics are computed independently for each task at $k \in \{1, 5, 10\}$ and macro-averaged across all test queries.

Let G_q denote the non-empty set of relevant documents for query q , and let $d_{q,i}$ denote the document at rank i after duplicate document identifiers have been removed. Relevance is binary:

$$\text{rel}_{q,i} = \begin{cases} 1, & d_{q,i} \in G_q, \\ 0, & d_{q,i} \notin G_q. \end{cases} \quad (2)$$

Discounted cumulative gain at cutoff k is defined as

$$\text{DCG}_q @k = \sum_{i=1}^k \frac{\text{rel}_{q,i}}{\log_2(i+1)}. \quad (3)$$

For binary relevance, the ideal ranking contains $\min(k, |G_q|)$ relevant documents. Therefore,

$$\text{IDCG}_q @k = \sum_{i=1}^{\min(k, |G_q|)} \frac{1}{\log_2(i+1)}, \quad (4)$$

and normalized discounted cumulative gain is

$$\text{nDCG}_q @k = \frac{\text{DCG}_q @k}{\text{IDCG}_q @k}. \quad (5)$$

Let r_q denote the rank of the first relevant document. The truncated reciprocal-rank and Hit@ k metrics are

$$\text{RR}_q @k = \begin{cases} 1/r_q, & r_q \leq k, \\ 0, & r_q > k, \end{cases} \quad (6)$$

$$\text{Hit}_q @k = \begin{cases} 1, & r_q \leq k, \\ 0, & r_q > k. \end{cases} \quad (7)$$

For any query-level metric $m_q @k$, the reported task-level score is the macro-average over the complete test-query set Q :

$$M @k = \frac{1}{|Q|} \sum_{q \in Q} m_q @k. \quad (8)$$

Thus, $\text{MRR} @k$ is the macro-average of $\text{RR}_q @k$, while $\text{Hit} @k$ is the proportion of test queries with at least one relevant document among the top- k retrieved results.

F Prompt Used for Generative Candidate Scoring

For LLM-based candidate scoring, each candidate disease was evaluated independently using the following prompt template. The same semantic prompt was supplied through each model's native chat template.

System instruction

You are evaluating the textual compatibility between a candidate disorder description and a set of patient phenotypes. Use ONLY the information provided in the phenotype list and the candidate disorder description. Do NOT use external medical knowledge.

Patient phenotypes

- Increased lower facial height
- Laryngeal stenosis
- Long philtrum

Candidate disorder description

Autosomal dominant prognathism. Malocclusion in which the mandible is anterior to the maxilla, as reflected by the relationship of the first permanent molars.

Task

Assign an integer compatibility score from 0 to 100 indicating how well the candidate disorder description supports the complete phenotype set.

Scoring guideline

- 0–19: incompatible
- 20–39: weak overlap
- 40–59: partial match
- 60–79: strong match
- 80–100: near-complete match

Output rules

Output exactly one JSON object containing only the score. Do not include explanations, Markdown, additional fields, or repeated output.

```
{"score": 0}
```

G Additional Evaluation Metrics

Model	nDCG@k								
	Gene→Def			Phen→Def			Dis→Def		
	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.079	0.083	0.087	0.265	0.361	0.381	0.200	0.262	0.275
MiniLM-L6-v2	0.158	0.244	0.263	0.500	0.609	0.629	0.224	0.296	0.314
BioLORD-0.1B‡	0.311	0.428	0.455	0.452	0.639	0.668	0.389	0.505	0.532
BioLORD-0.1B†	0.364	0.489	0.513	0.767	0.862	0.869	0.425	0.540	0.562
BioLORD-0.1B	0.167	0.268	0.290	0.693	0.795	0.807	0.309	0.402	0.426
MedEmbed-0.1B	0.320	0.424	0.440	0.600	0.722	0.738	0.350	0.433	0.454
SapBERT-0.1B	0.510	0.654	0.670	0.699	0.806	0.818	0.353	0.452	0.476
Qwen3-Embed-0.6B‡	0.348	0.476	0.499	0.390	0.541	0.572	0.335	0.460	0.488
Qwen3-Embed-0.6B†	0.480	0.597	0.615	0.762	0.855	0.864	0.401	0.515	0.538
Qwen3-Embed-0.6B	0.220	0.305	0.325	0.475	0.586	0.605	0.174	0.232	0.247
Qwen3-Embed-4B	0.378	0.501	0.526	0.476	0.592	0.616	0.214	0.306	0.330
OpenAI (text-embedding-3-large)	0.619	0.730	0.747	0.751	0.851	0.860	0.391	0.493	0.516

Table 6: Tier 1 ontology grounding tasks evaluated using nDCG@k.

Model	MRR@k								
	Gene→Def			Phen→Def			Dis→Def		
	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.079	0.082	0.083	0.265	0.333	0.341	0.200	0.244	0.250
MiniLM-L6-v2	0.158	0.219	0.227	0.500	0.579	0.587	0.224	0.275	0.282
BioLORD-0.1B‡	0.311	0.392	0.404	0.452	0.586	0.598	0.389	0.472	0.483
BioLORD-0.1B†	0.364	0.454	0.464	0.767	0.836	0.839	0.425	0.507	0.516
BioLORD-0.1B	0.167	0.237	0.246	0.693	0.766	0.771	0.309	0.375	0.385
MedEmbed-0.1B	0.320	0.395	0.402	0.600	0.688	0.695	0.350	0.409	0.418
SapBERT-0.1B	0.510	0.614	0.620	0.699	0.777	0.782	0.353	0.423	0.433
Qwen3-Embed-0.6B‡	0.348	0.438	0.447	0.390	0.497	0.509	0.335	0.423	0.435
Qwen3-Embed-0.6B†	0.480	0.565	0.573	0.762	0.830	0.834	0.401	0.482	0.491
Qwen3-Embed-0.6B	0.220	0.282	0.290	0.475	0.554	0.562	0.174	0.215	0.221
Qwen3-Embed-4B	0.378	0.466	0.477	0.476	0.558	0.568	0.214	0.279	0.289
OpenAI (text-embedding-3-large)	0.619	0.699	0.706	0.751	0.825	0.829	0.391	0.464	0.473

Table 7: Tier 1 ontology grounding tasks evaluated using MRR@k.

Model	Hit@k								
	Gene→Def			Phen→Def			Dis→Def		
	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.079	0.088	0.100	0.265	0.448	0.508	0.200	0.315	0.357
MiniLM-L6-v2	0.158	0.318	0.378	0.500	0.700	0.761	0.224	0.360	0.417
BioLORD-0.1B‡	0.311	0.534	0.619	0.452	0.797	0.886	0.389	0.605	0.686
BioLORD-0.1B†	0.364	0.592	0.666	0.767	0.937	0.958	0.425	0.639	0.706
BioLORD-0.1B	0.167	0.362	0.429	0.693	0.879	0.916	0.309	0.485	0.559
MedEmbed-0.1B	0.320	0.508	0.559	0.600	0.821	0.873	0.350	0.505	0.569
SapBERT-0.1B	0.510	0.775	0.821	0.699	0.892	0.929	0.353	0.538	0.612
Qwen3-Embed-0.6B‡	0.348	0.592	0.664	0.390	0.676	0.771	0.335	0.569	0.656
Qwen3-Embed-0.6B†	0.480	0.689	0.745	0.762	0.929	0.955	0.401	0.615	0.684
Qwen3-Embed-0.6B	0.220	0.374	0.436	0.475	0.681	0.741	0.174	0.283	0.329
Qwen3-Embed-4B	0.378	0.606	0.680	0.476	0.692	0.768	0.214	0.386	0.461
OpenAI (text-embedding-3-large)	0.619	0.824	0.876	0.751	0.928	0.956	0.391	0.582	0.652

Table 8: Tier 1 ontology grounding tasks evaluated using Hit@k.

Model	nDCG@k											
	Dis→Gene			Dis→Phen			Phen→Dis			Phen→Gene		
	@1	@5	@10	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.076	0.050	0.054	0.084	0.066	0.070	0.099	0.099	0.103	0.048	0.065	0.071
MiniLM-L6-v2	0.079	0.055	0.058	0.120	0.098	0.106	0.150	0.143	0.153	0.058	0.095	0.109
BioLORD-0.1B‡	0.112	0.083	0.089	0.158	0.142	0.160	0.034	0.119	0.166	0.104	0.165	0.191
BioLORD-0.1B†	0.061	0.043	0.048	0.130	0.104	0.112	0.151	0.159	0.175	0.054	0.078	0.091
BioLORD-0.1B	0.066	0.053	0.058	0.099	0.083	0.091	0.158	0.161	0.178	0.060	0.094	0.105
MedEmbed-0.1B	0.095	0.066	0.068	0.148	0.120	0.130	0.155	0.157	0.169	0.079	0.118	0.134
SapBERT-0.1B	0.096	0.066	0.069	0.104	0.080	0.088	0.142	0.148	0.159	0.068	0.101	0.113
Qwen3-Embed-0.6B‡	0.137	0.102	0.110	0.185	0.165	0.184	0.224	0.254	0.283	0.143	0.236	0.269
Qwen3-Embed-0.6B†	0.064	0.041	0.045	0.124	0.103	0.112	0.131	0.142	0.158	0.050	0.070	0.076
Qwen3-Embed-0.6B	0.089	0.060	0.063	0.104	0.084	0.089	0.139	0.139	0.151	0.066	0.091	0.106
Qwen3-Embed-4B	0.105	0.070	0.073	0.144	0.112	0.119	0.169	0.174	0.188	0.076	0.115	0.129
OpenAI (text-embedding-3-large)	0.092	0.065	0.069	0.140	0.110	0.116	0.176	0.181	0.195	0.084	0.121	0.136

Table 9: Tier-2 ontology relation tasks evaluated using nDCG@k.

Model	MRR@k											
	Dis→Gene			Dis→Phen			Phen→Dis			Phen→Gene		
	@1	@5	@10	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.076	0.105	0.114	0.084	0.120	0.128	0.099	0.143	0.150	0.048	0.060	0.062
MiniLM-L6-v2	0.079	0.112	0.120	0.120	0.175	0.187	0.150	0.202	0.214	0.058	0.084	0.090
BioLORD-0.1B‡	0.112	0.164	0.177	0.158	0.249	0.269	0.034	0.125	0.150	0.104	0.146	0.156
BioLORD-0.1B†	0.061	0.090	0.099	0.130	0.189	0.202	0.151	0.219	0.232	0.054	0.070	0.075
BioLORD-0.1B	0.066	0.103	0.113	0.099	0.152	0.163	0.158	0.228	0.241	0.060	0.084	0.088
MedEmbed-0.1B	0.095	0.134	0.143	0.148	0.211	0.225	0.155	0.217	0.229	0.079	0.106	0.113
SapBERT-0.1B	0.096	0.131	0.141	0.104	0.149	0.160	0.142	0.206	0.217	0.068	0.091	0.096
Qwen3-Embed-0.6B‡	0.137	0.195	0.210	0.185	0.287	0.306	0.224	0.324	0.341	0.143	0.208	0.222
Qwen3-Embed-0.6B†	0.064	0.089	0.097	0.124	0.182	0.195	0.131	0.193	0.206	0.050	0.064	0.066
Qwen3-Embed-0.6B	0.089	0.123	0.132	0.104	0.149	0.158	0.139	0.198	0.210	0.066	0.083	0.089
Qwen3-Embed-4B	0.105	0.142	0.152	0.144	0.204	0.215	0.169	0.238	0.251	0.076	0.103	0.109
OpenAI (text-embedding-3-large)	0.092	0.132	0.141	0.140	0.198	0.209	0.176	0.247	0.259	0.084	0.110	0.116

Table 10: Tier-2 ontology relation tasks evaluated using MRR@k.

Model	Hit@k											
	Dis→Gene			Dis→Phen			Phen→Dis			Phen→Gene		
	@1	@5	@10	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.076	0.160	0.232	0.084	0.184	0.245	0.099	0.218	0.277	0.048	0.081	0.100
MiniLM-L6-v2	0.079	0.172	0.236	0.120	0.273	0.369	0.150	0.295	0.379	0.058	0.128	0.173
BioLORD-0.1B‡	0.112	0.258	0.349	0.158	0.419	0.570	0.034	0.342	0.524	0.104	0.225	0.306
BioLORD-0.1B†	0.061	0.145	0.213	0.130	0.298	0.397	0.151	0.341	0.435	0.054	0.104	0.143
BioLORD-0.1B	0.066	0.171	0.248	0.099	0.248	0.334	0.158	0.354	0.449	0.060	0.124	0.159
MedEmbed-0.1B	0.095	0.203	0.274	0.148	0.326	0.429	0.155	0.328	0.415	0.079	0.155	0.204
SapBERT-0.1B	0.096	0.197	0.272	0.104	0.232	0.314	0.142	0.319	0.401	0.068	0.131	0.169
Qwen3-Embed-0.6B‡	0.137	0.303	0.410	0.185	0.470	0.611	0.224	0.498	0.622	0.143	0.319	0.422
Qwen3-Embed-0.6B†	0.064	0.140	0.202	0.124	0.286	0.378	0.131	0.302	0.398	0.050	0.088	0.106
Qwen3-Embed-0.6B	0.089	0.186	0.255	0.104	0.228	0.298	0.139	0.305	0.391	0.066	0.117	0.163
Qwen3-Embed-4B	0.105	0.211	0.288	0.144	0.308	0.395	0.169	0.360	0.456	0.076	0.149	0.192
OpenAI (text-embedding-3-large)	0.092	0.204	0.277	0.140	0.302	0.387	0.176	0.372	0.466	0.084	0.167	0.202

Table 11: Tier-2 ontology relation tasks evaluated using Hit@k.

Model	PhenTriplet→Dis								
	nDCG@ <i>k</i>			MRR@ <i>k</i>			Hit@ <i>k</i>		
	@1	@5	@10	@1	@5	@10	@1	@5	@10
BM25	0.066	0.093	0.107	0.066	0.103	0.110	0.066	0.172	0.230
MiniLM-L6-v2	0.072	0.106	0.123	0.072	0.110	0.119	0.072	0.180	0.246
BioLORD-0.1B‡	0.085	0.196	0.244	0.085	0.185	0.207	0.085	0.368	0.527
BioLORD-0.1B†	0.096	0.146	0.171	0.096	0.151	0.163	0.096	0.247	0.340
BioLORD-0.1B	0.082	0.126	0.151	0.082	0.131	0.143	0.082	0.224	0.312
MedEmbed-0.1B	0.083	0.122	0.143	0.083	0.123	0.134	0.083	0.202	0.281
SapBERT-0.1B	0.076	0.111	0.131	0.076	0.116	0.127	0.076	0.194	0.272
Qwen3-Embed-0.6B‡	0.183	0.275	0.313	0.183	0.281	0.297	0.183	0.449	0.572
Qwen3-Embed-0.6B†	0.083	0.134	0.158	0.083	0.135	0.147	0.083	0.231	0.321
Qwen3-Embed-0.6B	0.058	0.089	0.109	0.058	0.092	0.102	0.058	0.157	0.234
Qwen3-Embed-4B	0.028	0.060	0.080	0.028	0.061	0.071	0.028	0.123	0.202
OpenAI (text-embedding-3-large)	0.097	0.143	0.169	0.097	0.149	0.160	0.097	0.244	0.331

Table 12: Performance on the Tier-3 PhenTriplet→Dis retrieval task. We report nDCG, MRR, and Hit at $k \in \{1, 5, 10\}$. The † symbol identifies models fine-tuned using Tier 1 supervision, while ‡ identifies models jointly fine-tuned using supervision from all three tiers.

H Error Analysis (Tier 3)

Failure-mode analysis. We analyzed Tier-3 queries for which the embedding model did not rank the ground-truth disease first. For each failure, we computed phenotype overlap between the query and the predicted disease (Ov_{pred}), the rank of the ground-truth disease under the Phenomizer-style Resnik reference (r_{Resnik}), and mean query phenotype information content ($\overline{IC}_q$). Ontology recovery was defined as $r_{Resnik} \leq 10$; the low-information threshold was $\overline{IC}_q \leq 5.949$, the 25th percentile among embedding failures.

Failures were assigned sequentially to mutually exclusive categories: *generic-bias* if $\overline{IC}_q \leq 5.949$; *related-disease confusion* if $Ov_{pred} = 2$ and $r_{Resnik} \leq 10$; *aggregation failure* if $Ov_{pred} \leq 1$ and $r_{Resnik} \leq 10$; and *semantic drift* otherwise.

Interpretation. Tier-3 queries are phenotype triplets. Predictions matching two phenotypes are treated as near misses; those matching at most one are aggregation failures only when the ontology reference retrieves the true disease in the top 10, and semantic drift otherwise. Low-information is an operational query-specificity category, not a causal explanation.

Mode	QID	Phenotypes	Predicted disease (embedding model)	True disease	$r_{Phenomizer}$	Ov_{pred}	Ov_{true}
Aggregation failure	Q2599	Sloping forehead; 2-4 toe cutaneous syndactyly; Chiari type II malformation	Crossed polysyndactyly	Lathosterolosis	1	0	3
Aggregation failure	Q4636	Finger syndactyly; Ungual fibroma; Glue ear	Crossed polysyndactyly	Choroidal atrophy-alopecia syndrome	1	1	3
Aggregation failure	Q5613	Long philtrum; Conical incisor; Hypophosphaturia	Hypophosphatemic rickets, autosomal recessive, 2	Global developmental delay-osteopenia-ectodermal defect syndrome	1	0	3
Related disease confusion	Q4072	Flat face; Sacral dimple; Prominent fingertip pads	PDE4D haploinsufficiency syndrome	Vulto-van Silfout-de Vries syndrome	1	2	3
Related disease confusion	Q1943	Short philtrum; Prominent fingertip pads; Scoliosis	1q21.1 microdeletion syndrome	Global developmental delay-visual anomalies-progressive cerebellar atrophy-truncal hypotonia syndrome	1	2	3
Related disease confusion	Q2002	Midface retrusion; Scoliosis; Radial deviation of finger	Autosomal dominant Robinow syndrome	Acrofacial dysostosis 1, Nager type	1	2	3
Generic bias	Q18	Intellectual disability; Tremor; Tapered finger	Developmental delay and seizures with or without movement abnormalities	Agenesis of the corpus callosum with peripheral neuropathy	1	2	3
Generic bias	Q976	Abnormal cardiovascular system morphology; Long philtrum; Frontal bossing	1q21.1 microdeletion syndrome	Doors syndrome	1	2	3
Generic bias	Q730	Seizure; Splenomegaly; Pto-sis	Sialidosis type 2	Proteus syndrome, somatic	1	2	3
Semantic drift	Q6075	Pto-sis; Paraplegia; Brain atrophy	Primary lateral sclerosis	Oculopharyngodistal myopathy 1	12	0	3
Semantic drift	Q651	Brachydactyly; Limitation of joint mobility; Abnormal metaphysis morphology	Metaphyseal dysostosis-intellectual disability-conductive deafness syndrome	Thanatophoric dysplasia type 2	12	0	3
Semantic drift	Q2375	Generalized-onset seizure; Lower limb spasticity; Status epilepticus	Epileptic encephalopathy, early infantile, 6 (Dravet syndrome)	Multiple congenital anomalies-hypotonia-seizures syndrome 2	13	1	3

Table 13: Representative Tier-3 embedding failures grouped by rule-based error mode. The embedding model frequently retrieves partially compatible disorders despite full phenotype coverage by the ground-truth disease, while the Phenomizer baseline often resolves the same queries through ontology-based phenotype aggregation. QID denotes the query identifier. $r_{Phenomizer}$ denotes the ranking position assigned to the ground-truth disease by the ontology-aware Phenomizer baseline. Ov_{pred} and Ov_{true} denote phenotype overlap counts between the query phenotype set and the predicted or ground-truth disease, respectively.